Temperature-switchable thermal barriers against thermal runaway propagation

A one-dimensional finite-volume study

Aditya Meenakshisundaram

github.com/adityameenak/switchable-barrier-runaway

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Abstract

A thermal barrier between lithium-ion cells faces a conflict. A conductive barrier spreads heat to the cooling system in normal operation but carries a runaway cell's heat to its neighbours. An insulating barrier blocks propagation but traps heat during normal use. A barrier whose conductivity drops sharply above a switch temperature could, in principle, do both. This report quantifies that tradeoff with a one-dimensional transient finite-volume model of a five-cell stack, using single-step Arrhenius self-heating, a smooth reversible switch law $k(T)$ and convective cooling at the stack ends. With illustrative parameters, a 2 mm switchable barrier holds the normal-operation peak temperature at the conductive barrier's 38.0 °C (against 64.5 °C for an aerogel-like insulator) while confining runaway to the trigger cell, whereas a conductive barrier lets runaway spread to all five cells within about a minute. A sweep of 361 barrier designs shows that containment is governed by a single quantity, the hot-state barrier conductance k_{off}/L, which must stay below about 32 $\mathrm{W\,m^{-2}\,K^{-1}}$ for the chosen end cooling. A switch-temperature sweep finds a window of about 38–175 °C in which the switch both cools and protects. The solver conserves energy to round-off and is verified against analytical, ODE-solver and convergence tests. The convergence study also showed that onset-fitted Arrhenius kinetics, extrapolated to burning-cell temperatures, produce a reaction front too thin for any practical mesh to resolve; a rate ceiling restores mesh convergence. The model is then extended with tabulated $k(T)$, latent heat and a delayed, one-way switch, and six real barrier materials are compared using datasheet and literature properties. Static insulators either contain runaway only with a large normal-use penalty (silica aerogel, 79 °C at its 2.25 mm containment thickness; ceramic fibre, 66 °C at 4.25 mm) or need thick layers (mica, 6.25 mm), and a conductive gap pad and a paraffin/graphite phase-change composite do not contain it at any thickness up to 10 mm. A published graphene/microsphere thermal switch contains runaway at 3.5 mm while keeping the normal-use peak within 1 K of the ideal conductor, and is the best of the six under every sensitivity case tested. Cell and kinetic parameters remain order-of-magnitude placeholders, so the conclusions are qualitative.

1 Introduction

Thermal runaway in a lithium-ion cell is a self-accelerating sequence of exothermic decomposition reactions that can drive the cell to several hundred degrees and lead to fire [1–3]. In a module, the heat released by one failing cell can trigger runaway in its neighbours, as multi-cell abuse experiments have shown repeatedly [4, 5]. Preventing this cell-to-cell propagation is a central goal of battery-pack safety design. The usual approach places a thermal barrier between cells, which creates a design conflict:

- a *conductive* barrier helps move the heat cells generate in normal operation toward the cooling system, but also passes a runaway cell's heat directly to its neighbour;
- an *insulating* barrier, for example an aerogel sheet, blocks propagation but raises cell temperatures in everyday use, which accelerates ageing.

Temperature-responsive barriers and thermal switches, whose conductivity changes with temperature, are an active research topic as a way out of this conflict [6, 7].

Related work. Thermal abuse of single cells is commonly modelled by coupling heat conduction to Arrhenius kinetics for a set of decomposition reactions, an approach established for cylindrical and prismatic cells [8], applied to abuse behaviour of high-power cells [9], extended to three dimensions [10] and combined with experiments [11]. Propagation models extend this to packs and modules, with lumped [12] and three-dimensional descriptions [13] and analyses of the heat-transfer paths between cells [14]. On the mitigation side, studies have examined phase-change composites [15], aerogel-based insulation [16], model-based optimisation of heat-absorbing interstitial barriers [17] and polymer barrier materials [18]. These barriers are static: their conductivity does not respond to the operating state. Thermal switches, devices whose conductance changes with temperature or another stimulus, are reviewed in general in [19] and for batteries in [7], and a temperature-responsive regulator for battery modules has been demonstrated experimentally [6].

This work. This report complements that work with a deliberately minimal model that isolates the conductivity-switching function of a barrier, and maps which switch properties are needed for containment without penalising normal operation. It asks three questions:

1. Does an idealised switchable barrier deliver both the conductor's normal-use temperature and the insulator's containment?
2. What hot-state conductivity and thickness does it need?
3. How should its switch temperature be chosen?
4. Which of six real barrier materials, with properties taken from datasheets and the literature, best balances containment against normal-use temperature?

The emphasis is on a transparent, verified model rather than quantitative prediction. The cell and kinetic parameters are illustrative, order-of-magnitude values and have not been fitted to any particular cell. The barrier properties in the material comparison (Section 5.5) are sourced, with every assumed value flagged.

2 Model

2.1 Geometry and governing equations

The stack is five cells of thickness $L_c = 10\,\mathrm{mm}$ separated by four barriers of thickness L (default $2\,\mathrm{mm}$), modelled in one dimension through the stack thickness. Temperature obeys transient conduction with a temperature-dependent conductivity and volumetric sources,

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k(T)\,\frac{\partial T}{\partial x}\right) + q_{\mathrm{rxn}} + q_{\mathrm{gen}}, \tag{1}$$

with convective boundary conditions at both ends of the stack,

$$-k\,\frac{\partial T}{\partial x}\bigg|_{x=0} = h\,(T_\infty - T), \qquad k\,\frac{\partial T}{\partial x}\bigg|_{x=L_{\mathrm{stack}}} = h\,(T_\infty - T). \tag{2}$$

2.2 Cell self-heating

Each cell node carries a lumped reaction progress $\alpha \in [0, 1]$ governed by single-step kinetics, a simplification of the multi-reaction schemes used in detailed abuse models [8–10]:

$$\frac{\mathrm{d}\alpha}{\mathrm{d}t} = k_{\mathrm{eff}}(T)\,(1-\alpha), \qquad q_{\mathrm{rxn}} = \Delta H_v \frac{\mathrm{d}\alpha}{\mathrm{d}t}, \qquad \Delta H_v = \rho c_p\,\Delta T_{\mathrm{ad}}, \tag{3}$$

so that a cell reacting adiabatically heats by exactly ΔT_{ad}. The rate constant combines an Arrhenius term with a smooth ceiling,

$$k_{\mathrm{eff}} = \frac{k_{\mathrm{arr}}\,k_{\max}}{k_{\mathrm{arr}} + k_{\max}}, \qquad k_{\mathrm{arr}} = A\,e^{-E_a/RT}. \tag{4}$$

With $E_a = 169\,\mathrm{kJ\,mol^{-1}}$ and $A = 7.3 \times 10^{16}\,\mathrm{s^{-1}}$, k_{arr} is about $10^{-4}\,\mathrm{s^{-1}}$ at 150 °C (slow onset) and about $1\,\mathrm{s^{-1}}$ at 250 °C. The ceiling $k_{\max} = 1\,\mathrm{s^{-1}}$ leaves onset unchanged; its purpose is explained in Section 4.1.

2.3 Barrier designs

Four designs are compared: no barrier (cells in direct contact); a static conductor ($k = 5\,\mathrm{W\,m^{-1}\,K^{-1}}$); a static, aerogel-like insulator ($k = 0.03\,\mathrm{W\,m^{-1}\,K^{-1}}$); and a switchable barrier that conducts below a switch temperature T_{sw} and insulates above it. The switch law is a logistic blend in $\ln k$,

$$\ln k(T) = \ln k_{\mathrm{off}} + \frac{\ln(k_{\mathrm{on}}/k_{\mathrm{off}})}{1 + e^{(T-T_{\mathrm{sw}})/w}}, \tag{5}$$

with $k_{\mathrm{on}} = 5\,\mathrm{W\,m^{-1}\,K^{-1}}$, $k_{\mathrm{off}} = 0.05\,\mathrm{W\,m^{-1}\,K^{-1}}$, $T_{\mathrm{sw}} = 120\,^{\circ}\mathrm{C}$ and $w = 5\,\mathrm{K}$. The switch is reversible and local: every barrier node switches on its own temperature, so a barrier with a steep internal gradient can be insulating on its hot side and conducting on its cold side. Blending $\ln k$ rather than k matters for a 100× switch ratio. A linear blend leaves k several times k_{off} until about 20 K above T_{sw}, so the barrier would barely insulate near its nominal switch point. All barriers share the same volumetric heat capacity, which isolates the effect of conductivity.

2.4 Real barrier materials

Comparing real materials needs three extensions of the barrier model. Each reduces to the idealised designs when switched off, and all earlier results are unchanged.

- **Tabulated conductivity.** $k(T)$ is interpolated from published points, linearly in $\ln k$, and held constant beyond the first and last point.
- **Latent heat.** A phase-change barrier stores enthalpy

$$H(T) = \rho c_p (T - T_{\mathrm{ref}}) + \rho L\,\Phi\!\left(\frac{T - T_m}{\sigma_m}\right), \tag{6}$$

  with latent heat L released over a Gaussian melting range (Φ is the normal cumulative distribution, $\sigma_m = 3\,\mathrm{K}$).
- **Switch dynamics.** The conducting fraction s of a switch (the exponent in Eq. (5)) relaxes toward its equilibrium value $s_{\mathrm{eq}}(T)$ with time constant τ, $\mathrm{d}s/\mathrm{d}t = (s_{\mathrm{eq}} - s)/\tau$. A one-way switch can only move toward the insulating state, $s^{n+1} = \min(s^n, \cdot)$.

Six candidates were chosen to cover the main barrier mechanisms (Table 1): three static insulators (silica aerogel blanket, phlogopite mica paper, alumina–silica ceramic fibre), a static conductor (silicone gap pad), a phase-change composite (paraffin with 10 wt% expanded graphite) and a thermal switch (the graphene/expandable-microsphere regulator of Wang et al. [6]). Every value is taken from a manufacturer datasheet or a peer-reviewed paper, except where marked as assumed. The data and their provenance are kept in `runaway/candidates.py`, which also records how each material is mapped onto the model.

Table 1: Barrier candidates. k is through-plane, at the temperatures shown. [a]Assumed generic value for the material class (not in the source). [b]From a search summary of the source, not yet checked against the original.

Material (source)	k [$\mathrm{W\,m^{-1}K^{-1}}$] at T [°C]	ρ [$\mathrm{kg\,m^{-3}}$]	c_p [$\mathrm{J\,kg^{-1}K^{-1}}$]	Other
Silica aerogel [20]	0.020 @ 0, 0.028 @ 200, 0.046 @ 400, 0.089 @ 600	200	1000[a]	max. 650 °C
Mica paper [21]	0.23 @ 50	1900	850[a]	max. 700 °C
Ceramic fibre [22]	0.06 @ 260, 0.13 @ 538, 0.23 @ 816	128	1000[a]	max. 1177 °C
Silicone gap pad [23]	3.0[b] @ 25	3200[b]	1000[b]	max. 200 °C[b]
Paraffin/EG PCM [24]	9.99 @ 25[a]	1103	2000[a]	$L = 240\,\mathrm{kJ\,kg^{-1}}$, $T_m \approx 74$ °C
Graphene switch [6]	1.33 @ 25, 0.10 @ ∼100	1000[a]	1000[a]	switches within 30 s

Where the data do not determine a model input, the choice is stated and later varied (Table 5). The graphene switch is modelled with $k_{\mathrm{on}} = 1.33$, $k_{\mathrm{off}} = 0.10\,\mathrm{W\,m^{-1}K^{-1}}$, $T_{\mathrm{sw}} = 85$ °C and $w = 5$ K, so that k reaches about $0.11\,\mathrm{W\,m^{-1}K^{-1}}$ at 100 °C, and its 30 s response as a lag with $\tau = 10$ s. Expanding microspheres suggest a one-way switch, which is the default. The gap pad and the PCM are given constant k. Several of these products are not made for battery barriers: the aerogel and ceramic fibre data are for industrial blankets, and the PCM was developed for spacecraft electronics. The gap pad is rated only to 200 °C and the PCM was tested only to 85 °C, so their behaviour at runaway temperatures is an extrapolation.

2.5 Scenarios and parameters

Abuse. Cell 1 starts at 300 °C and everything else at 25 °C. A cell is counted as *propagated* when its volume-mean α exceeds 0.9.

Normal operation. Every cell generates a uniform $q_{\mathrm{gen}} = 2 \times 10^4\,\mathrm{W\,m^{-3}}$, the reaction is switched off (its rate is negligible below 70 °C), and Eq. (1) is solved to steady state. The quantity of interest is the peak cell temperature.

Parameters. Table 2 lists the parameters. Two were changed from the initial estimates during development. First, the end cooling coefficient was raised to $h = 100\,\mathrm{W\,m^{-2}\,K^{-1}}$, representing an end plate connected to a cooling system. With natural-convection values ($h \lesssim 30\,\mathrm{W\,m^{-2}\,K^{-1}}$) no barrier, not even the insulator, prevented propagation in this 1D model, because the stack ends are the only heat sink and the trigger cell's roughly $20\,\mathrm{MJ\,m^{-2}}$ must eventually leave somewhere. Second, the rate ceiling $k_{\max}$ was added (Section 4.1).

Table 2: Model parameters. All values are illustrative order-of-magnitude placeholders and have not been fitted to a specific cell or material.

Quantity	Value	Note
Cell k, ρ, c_p	$0.8\,\mathrm{W\,m^{-1}K^{-1}}$, $2500\,\mathrm{kg\,m^{-3}}$, $1000\,\mathrm{J\,kg^{-1}K^{-1}}$	through-plane, homogenised
E_a, A	$169\,\mathrm{kJ\,mol^{-1}}$, $7.3\times10^{16}\,\mathrm{s^{-1}}$	single-step kinetics
ΔT_{ad}, $k_{\max}$	$600\,\mathrm{K}$, $1\,\mathrm{s^{-1}}$	
Conductor / insulator k	5 / $0.03\,\mathrm{W\,m^{-1}K^{-1}}$	static
Switch k_{on}, k_{off}, T_{sw}, w	5 and $0.05\,\mathrm{W\,m^{-1}K^{-1}}$, $120\,^\circ\mathrm{C}$, $5\,\mathrm{K}$	reversible
Barrier ρc_p, thickness	$10^6\,\mathrm{J\,m^{-3}K^{-1}}$, $2\,\mathrm{mm}$	all designs
h, T_∞	$100\,\mathrm{W\,m^{-2}K^{-1}}$, $25\,^\circ\mathrm{C}$	both ends
q_{gen} (normal operation)	$2\times10^4\,\mathrm{W\,m^{-3}}$	cells only

3 Numerical method

Spatial discretisation. Each layer is meshed separately (40 volumes per cell, 12 per barrier), so every material interface lies on a control-volume face. The face conductance is the series resistance of the two adjoining half-volumes,

$$G_{i+1/2} = \left(\frac{\Delta x_i}{2k_i} + \frac{\Delta x_{i+1}}{2k_{i+1}} \right)^{-1}, \tag{7}$$

which is the distance-weighted harmonic mean of k [25]. It is exact for steady conduction through layers in series and handles the $27\times$ jump at an insulator–cell interface without special treatment. Boundary faces add $1/h$ in series.

Time integration. Reaction and conduction are coupled by first-order (Lie) operator splitting [26]. In the *reaction step*, Eq. (3) is linear in α for a frozen rate constant and has the exact solution

$$\alpha^{n+1} = 1 - (1-\alpha^n)\, e^{-k_{\mathrm{eff}}\Delta t}, \tag{8}$$

where k_{eff} is evaluated at a predicted mid-step temperature (an exponential midpoint rule). The released heat is added as $\Delta T_{\mathrm{ad}}\,(\alpha^{n+1}-\alpha^n)$. The update is stable for any Δt, keeps $0\le\alpha\le1$ and conserves energy exactly, which makes the stiff kinetics tractable. In the *conduction step*, backward Euler is used with $k(T)$ lagged from the start of the step (Picard refinement is available). The resulting tridiagonal system is solved with a banded direct solver.

Step control. The time step is

$$\Delta t = \min\left(\Delta t_{\max},\ \frac{\Delta T_{\mathrm{rxn,max}}}{\max_i \dot{T}_{\mathrm{rxn},i}} \right), \tag{9}$$

with $\Delta T_{\mathrm{rxn,max}} = 2\,\mathrm{K}$ and $\Delta t_{\max} = 0.5\,\mathrm{s}$. This is an accuracy limit rather than a stability limit: it shrinks the step only while some node is igniting.

Steady state. Normal operation is solved by pseudo-transient continuation: backward-Euler steps from ambient with a geometrically growing step. For the switchable barrier this follows the physical start-up path, so if self-heating trips the switch, the solution lands on the hot (insulating) branch.

Latent heat. With a phase-change barrier, the heat capacity in the backward-Euler step is replaced by the chord capacity $C^* = [H(T^{n+1}) - H(T^n)]/(T^{n+1} - T^n)$ from Eq. (6), and the step is iterated to a tolerance of 10^{-7} K. The stored-energy change then equals the enthalpy change exactly, so a node that crosses the whole melting range in one step cannot skip the latent heat, which an apparent-heat-capacity method evaluated at the old temperature would do.

Energy accounting. The solver tracks stored enthalpy, unreleased chemical energy and cumulative boundary loss. Because both sub-steps are conservative, the discrete energy balance closes to round-off. The largest relative residual in any run reported here is 3×10^{-10} (a phase-change barrier, set by the iteration tolerance); without latent heat it is below 10^{-11}.

4 Verification

The test suite runs automatically on every commit and checks the following.

1. **Energy conservation.** With insulated ends and no reaction, total thermal energy stays constant to a relative 10^{-12} over 2000 s, with the nonlinear switchable barrier in place. With the reaction on, thermal plus chemical energy is conserved, and with convection the budget closes to below 10^{-10}.
2. **Series conduction.** For a cell | barrier | cell stack held between two different ambient temperatures, the steady solution matches the analytical series-resistance profile to 10^{-8} K, for both 6× and 27× conductivity jumps, on a deliberately coarse mesh. For uniform generation in a single cell, the peak matches $T_\infty + qL_c/(2h) + qL_c^2/(8k)$.
3. **Reaction.** A single adiabatic reacting node ends at exactly $T_0 + \Delta T_{\mathrm{ad}}$. Its time to 50 % conversion matches a tight-tolerance implicit ODE solution (Radau IIA [27]) to within 1 %, with and without the rate ceiling.
4. **Convergence.** The cell-2 propagation time converges under mesh refinement with an observed order above 1.5, and under refinement of $\Delta T_{\mathrm{rxn,max}}$ (Fig. 1).
5. **Material extensions.** Tabulated $k(T)$ reproduces the published points. In an insulated stack with a phase-change barrier, enthalpy is conserved to 10^{-9} and the final uniform temperature satisfies the enthalpy balance including melting. A one-way switch heated past T_{sw} stays insulating after cooling, while a reversible one recovers, and a larger τ delays switching.

4.1 Why a rate ceiling is needed

The kinetic parameters describe the onset region (150–250 °C). Extrapolated to a burning cell near 900 °C, pure Arrhenius kinetics give $k_{\mathrm{arr}} \sim 10^9\,\mathrm{s}^{-1}$, which implies a reaction front inside the cell of order $\sqrt{\kappa/k_{\mathrm{arr}}} \sim 10^{-7}$ m, where $\kappa = k/\rho c_p$ is the thermal diffusivity. No practical mesh resolves this, so the front speed is set by the mesh rather than by the physics. The convergence test exposed the problem. Without the ceiling, the cell-2 propagation time drifts from 14.3 to 10.9, 11.1 and 8.8 s as the mesh is refined from 10 to 80 volumes per cell, with no sign of settling (Fig. 1).

Real high-temperature decomposition is limited by transport and multi-step chemistry, not by an ever-faster single step. The smooth ceiling in Eq. (4) with $k_{\mathrm{max}} = 1\,\mathrm{s}^{-1}$ makes a single cell burn over a few seconds, broadens the front to about $\sqrt{\kappa/k_{\mathrm{max}}} \approx 0.6$ mm and restores convergence. The default mesh is within 2 % of a 4× finer one, and halving $\Delta T_{\mathrm{rxn,max}}$ changes the result by 0.1 %. The contained cases are also converged: the peak temperature in cell 2 behind the switchable barrier varies by less than 0.1 K over a 100× range of Δt_{max}, and by about 1 K from 20 to 80 volumes per cell.

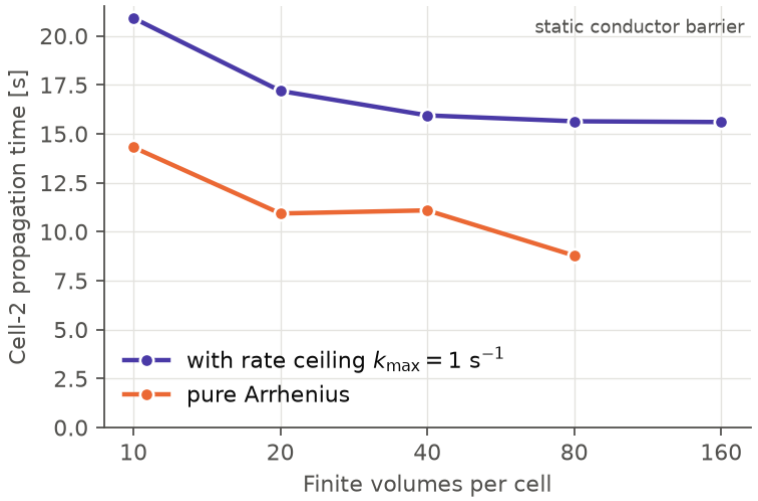


Figure 1: Mesh convergence of the cell-2 propagation time for the static-conductor stack. With pure Arrhenius kinetics the result does not converge, because the reaction front is unresolvably thin. With the rate ceiling it converges at roughly second order.

5 Results

5.1 The tradeoff

Table 3 and Fig. 2 summarise the four designs. Without a barrier or with the conductor, runaway spreads to every cell: each neighbour ignites 14–16 s after the previous one, and the whole stack has reacted within about one minute (Fig. 3). The insulator confines runaway to the trigger cell but raises the normal-operation peak by 26.5 K. The switchable barrier gives the conductor's 38.0 °C in normal use *and* the insulator's containment.

Table 3: Outcomes for the four designs (2 mm barriers where present).

Design	Normal-op. peak	Cells in runaway	Propagation times, cells 1–5 [s]
None (direct contact)	37.8 °C	5/5	2.3, 14.1, 28.2, 42.2, 56.1
Static conductor	38.0 °C	5/5	2.3, 15.9, 31.8, 47.8, 63.6
Static insulator	64.5 °C	1/5	2.3, –, –, –, –
Switchable	38.0 °C	1/5	2.3, –, –, –, –

The trigger cell peaks at 850–870 °C, slightly below $300 + \Delta T_{ad} = 900$ °C, because it loses heat while it reacts. Propagated cells settle near $25 + 600$ °C plus whatever preheating they received. Both values are consistent with the energy content of the model.

5.2 Mechanism of containment

Figure 4 shows how the switch works. The hot face of the first barrier switches off and holds a temperature drop of roughly 700 K across a few millimetres while cell 1 sheds its heat through the cooled end. The cold side of the same barrier stays below T_{sw} and keeps conducting, as do the downstream barriers, so cell 2 spreads whatever heat leaks through into cells 3–5. One consequence is easy to miss. Cell 2 reaches a higher local peak behind the switchable barrier (153 °C, 3 % conversion) than behind the static insulator (124 °C, 0.2 %), because only part of the switchable barrier's thickness is in the insulating state.

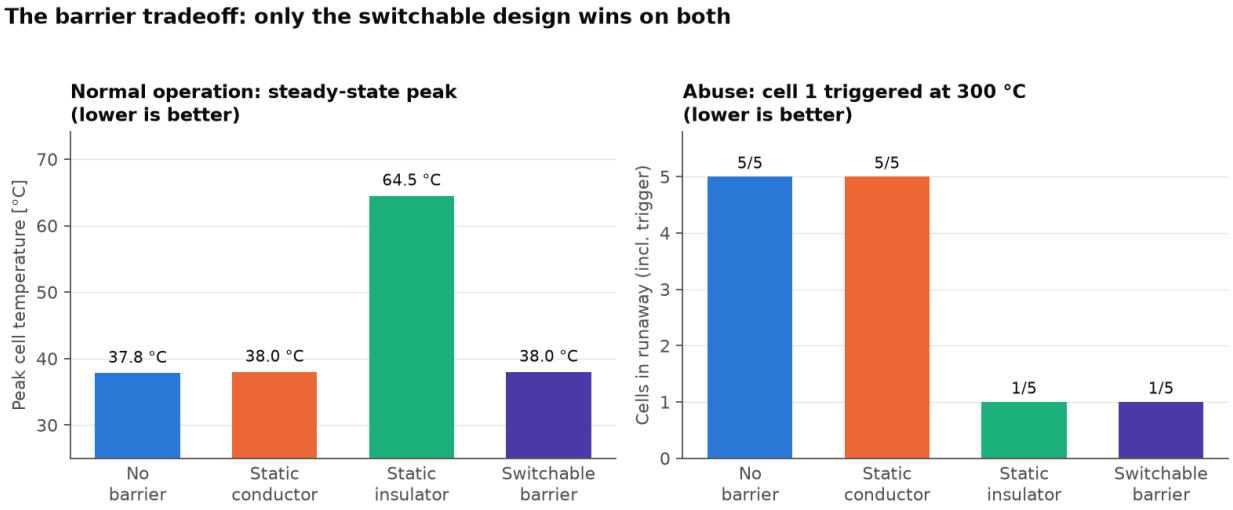


Figure 2: Normal-operation steady-state peak temperature (left) and number of cells in runaway after triggering cell 1 (right) for each design.

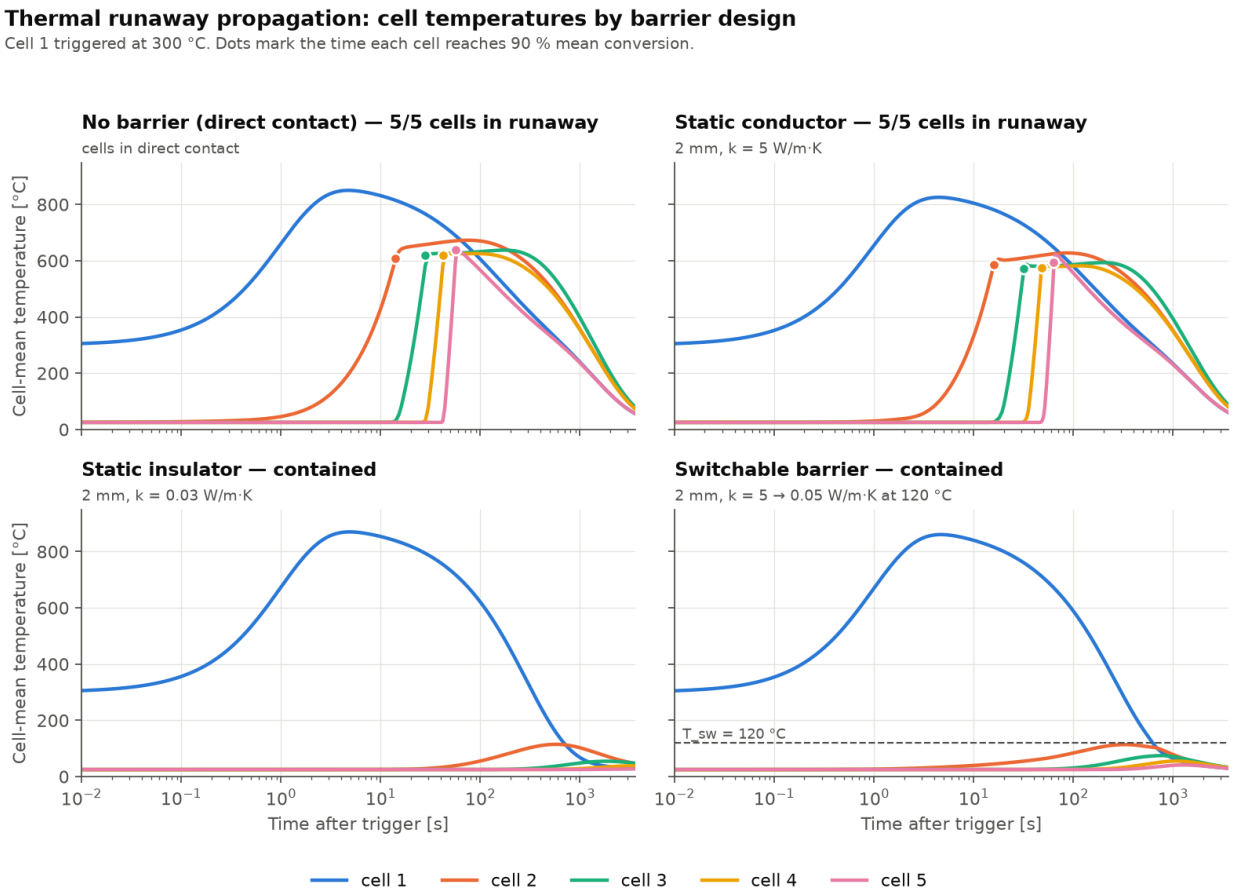


Figure 3: Cell-mean temperatures after triggering cell 1, one panel per barrier design. Dots mark the time each cell reaches 90 % mean conversion.

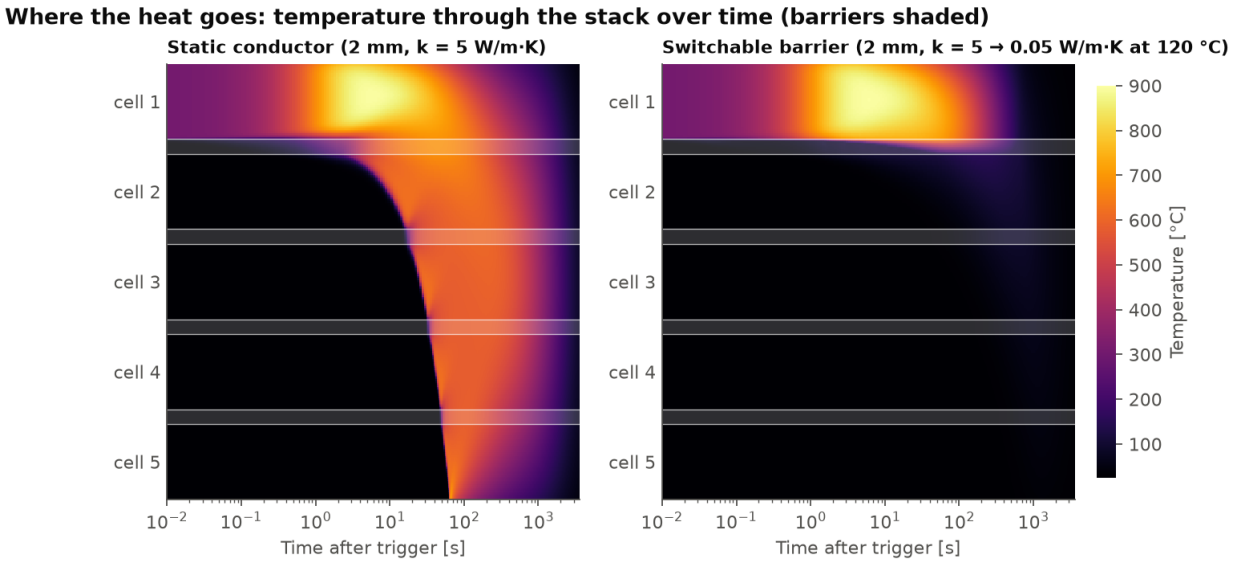


Figure 4: Temperature through the stack over time for the conductor (left) and the switchable barrier (right). Barriers are shaded.

5.3 Design map

Figure 5 varies the switchable barrier's thickness L (0.5–5 mm) and hot-state conductivity k_{off} (0.02–1 $\text{W m}^{-1}\text{K}^{-1}$) over 361 simulations. Two features stand out. First, the outcomes are binary: every run ends with either one or all five cells in runaway. Once cell 2 ignites, each later cell faces a trigger at least as severe, so the barrier must stop the first hand-off. Second, the boundary follows a line of constant hot-state conductance, $k_{\text{off}}/L \approx 32\,\text{W m}^{-2}\text{K}^{-1}$, about a third of the end-cooling coefficient. Physically, the runaway cell must lose heat to the cooled end faster than it leaks into its neighbour. The default design ($k_{\text{off}}/L = 25\,\text{W m}^{-2}\text{K}^{-1}$) lies about 20 % inside the boundary, a modest margin.

5.4 Choosing the switch temperature

Figure 6 sweeps T_{sw} from 25 to 250 °C. Below about 38 °C the barrier partly switches off during normal operation and the steady peak rises toward the insulator's. Above about 175 °C the barrier switches only after the face of cell 2 has passed self-heating onset, and runaway propagates. Between these limits the switch both cools and protects. The lower edge is set by the operating temperature plus a few switch widths; the upper edge is set by the kinetics.

5.5 Real barrier materials

Each candidate in Table 1 was run at thicknesses from 0.5 to 10 mm in 0.25 mm steps, in both the abuse and the normal-operation scenario. Two numbers summarise a material: the thinnest barrier that confines runaway to the trigger cell, and the normal-operation peak temperature at that thickness (Fig. 7, Table 4).

No candidate contains runaway at 2 mm. Even the aerogel, whose room-temperature k is below that of the idealised insulator, lets runaway through at 2 mm. Its conductivity triples between 200 and 600 °C, the range the barrier's hot face reaches, so its effective hot-state conductance at

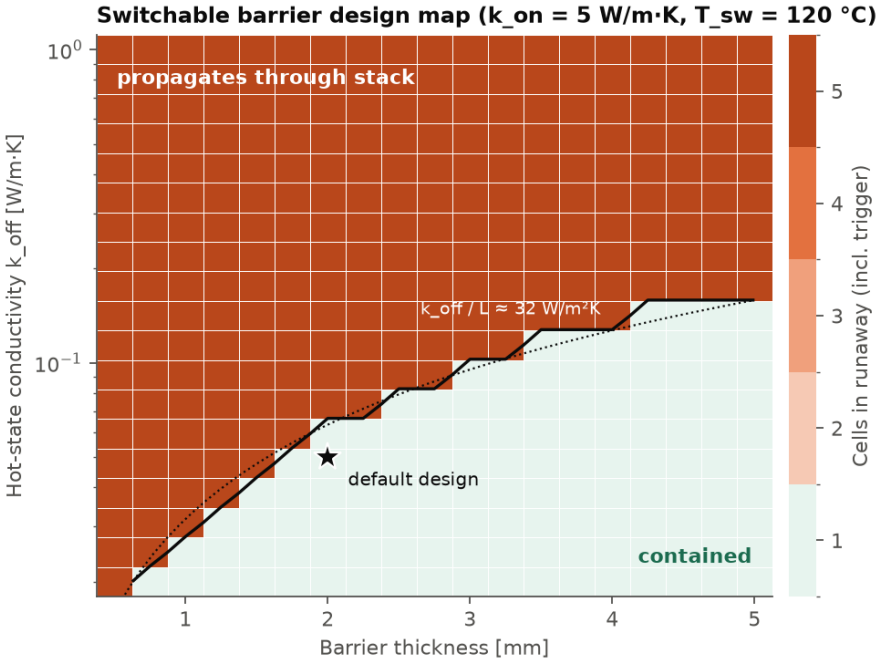


Figure 5: Cells in runaway over barrier thickness and hot-state conductivity ($k_{\mathrm{on}} = 5\,\mathrm{W\,m^{-1}K^{-1}}$, $T_{\mathrm{sw}} = 120\,^{\circ}\mathrm{C}$). The dotted line is constant k_{off}/L; the star marks the default design.

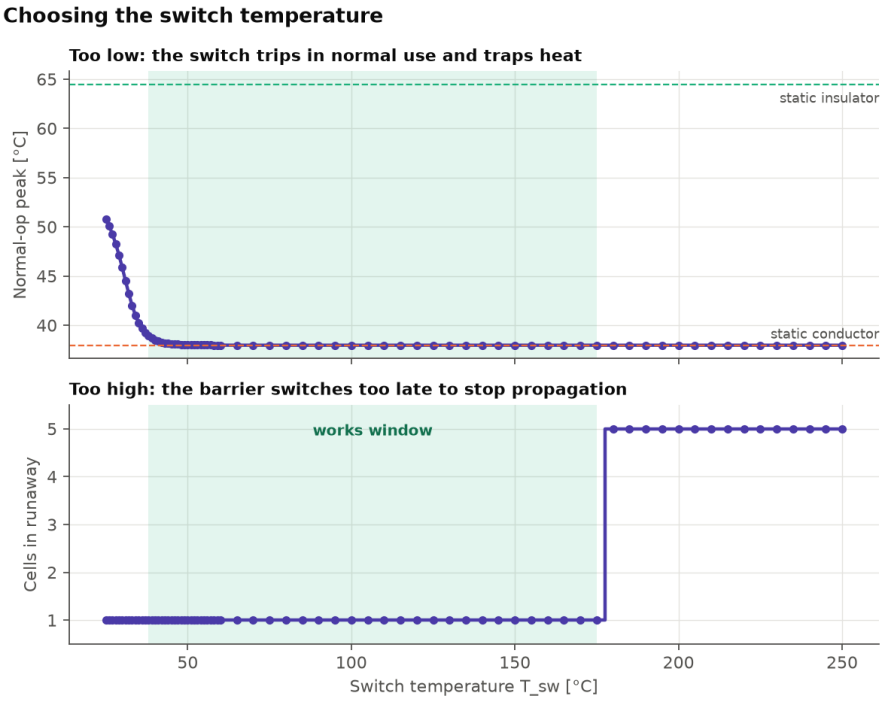


Figure 6: Normal-operation peak (top) and cells in runaway (bottom) as functions of the switch temperature. The shaded window both cools and protects.

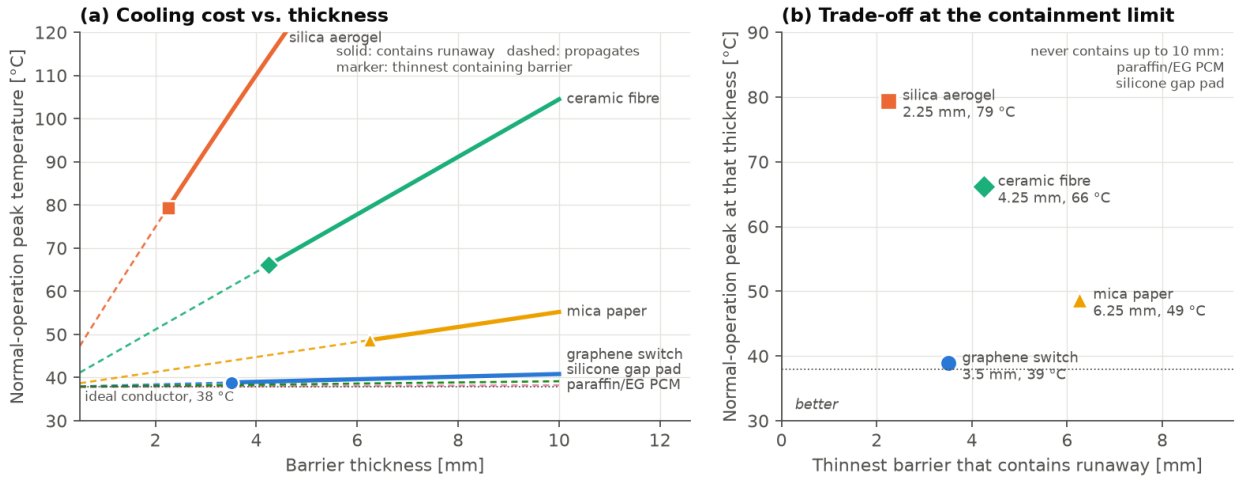


Figure 7: Real barrier materials. (a) Normal-operation peak temperature against barrier thickness; the line is solid where the barrier contains runaway and dashed where it propagates, and the marker shows the thinnest containing barrier. (b) Each material's thinnest containing barrier and its normal-operation peak at that thickness. The lower left is better. The dotted line is the ideal $5\,\mathrm{W\,m^{-1}K^{-1}}$ conductor at 2 mm.

Table 4: Material comparison. "Never" means runaway reached at least one neighbour at every thickness up to 10 mm. At 2 mm all six let runaway reach all five cells.

Material	Thinnest containing barrier [mm]	Normal peak there [°C]	Normal peak at 2 mm [°C]
Graphene switch	3.5	38.9	38.4
Silica aerogel	2.25	79.4	74.9
Ceramic fibre	4.25	66.1	51.1
Mica paper	6.25	48.7	41.3
Paraffin/EG PCM	never	–	37.9
Silicone gap pad	never	–	38.1

2 mm is about $40\,\mathrm{W\,m^{-2}K^{-1}}$, above the containment threshold of Section 5.3. What matters for containment is a barrier's conductivity at runaway temperatures, which datasheets often report only incompletely.

Static insulators trade thickness against normal-use temperature. Aerogel contains runaway with the thinnest layer, 2.25 mm, but raises the normal-operation peak by 41 K over the conductor. Ceramic fibre needs 4.25 mm and costs 28 K. Mica paper conducts heat about ten times better than aerogel, so its normal-use penalty is small (11 K), but it needs 6.25 mm before its conductance falls below the threshold. For all three the penalty grows steadily with thickness (Fig. 7a).

The conductor and the phase-change composite fail. The gap pad behaves like the idealised conductor and never contains runaway. The PCM also conducts well ($k \approx 10\,\mathrm{W\,m^{-1}K^{-1}}$), so it keeps cells cool, and its latent heat does slow propagation: at 4 mm, runaway reaches cell 5 after 148 s instead of 100 s without latent heat. But a 4 mm layer stores only about $1\,\mathrm{MJ\,m^{-2}}$ of latent heat, against about $15\,\mathrm{MJ\,m^{-2}}$ of reaction heat released by each cell, and once melted it passes heat on readily. Only at 9.5–10 mm does it hold runaway to two cells.

The thermal switch avoids the trade-off. The graphene switch contains runaway at 3.5 mm while keeping the normal-operation peak at 38.9 °C, within 1 K of the ideal conductor. Its hot-state conductance at that thickness, $0.10/0.0035 \approx 29\,\mathrm{W\,m^{-2}K^{-1}}$, is just below the threshold found with the idealised switch, so the design-map criterion carries over to a real material. Because its cold-state conductivity is only $1.33\,\mathrm{W\,m^{-1}K^{-1}}$, the normal-use penalty does grow with thickness, but slowly (41 °C at 10 mm).

Sensitivity. The switch data come from a paper abstract, so its density, heat capacity, switch temperature and response time are uncertain. Table 5 varies each one. Across these cases the thinnest containing switch ranges from 3.25 to 5 mm, with the normal-operation peak between 38.8 and 39.3 °C. The result is most sensitive to the hot-state conductivity, as the k_{off}/L criterion predicts, and to a slow response: with $\tau = 30\,\mathrm{s}$ the switch needs 4.5 mm. Reversibility makes no difference here, because the barrier next to the runaway cell does not cool below T_{sw} during the run. Doubling the switch's assumed heat capacity, or cutting the aerogel's by 30 %, changes its thinnest containing barrier by at most 0.25 mm.

Table 5: Sensitivity of the thinnest containing barrier to inputs the data do not pin down. Baseline: graphene switch 3.5 mm, aerogel 2.25 mm.

Case	Thinnest containing [mm]	Normal peak there [°C]
Switch, $T_{\mathrm{sw}} = 70\,°\mathrm{C}$	3.75	38.9
Switch, $T_{\mathrm{sw}} = 100\,°\mathrm{C}$	3.5	38.9
Switch, instantaneous ($\tau = 0$)	3.25	38.8
Switch, slow ($\tau = 30\,\mathrm{s}$)	4.5	39.2
Switch, reversible	3.5	38.9
Switch, $\rho c_p \times 2$	3.25	38.8
Switch, $k_{\mathrm{off}} \times 1.5$	5.0	39.3
Aerogel, $\rho c_p \times 0.7$	2.25	79.4
PCM without latent heat	never	–

Which material is best? Judged on both criteria together, the graphene switch is the best of the six. It is the only candidate that contains runaway at a practical thickness without a normal-use penalty, and it stays the best in every sensitivity case. Aerogel is the choice if thickness is the only constraint and the pack can remove the extra heat another way, for example with side cooling, which this 1D model cannot represent. Among commercially established products, mica paper is the best compromise if about 6 mm per gap and a normal-use peak near 49 °C are acceptable. The switch result rests on a laboratory material characterised here only by its published headline numbers, so measured $k(T)$, heat capacity and cycling data are needed before it can be recommended for a real pack.

6 Discussion and limitations

The model is deliberately minimal, and several simplifications bound what it can say.

- **One dimension.** Heat flows only through the stack thickness, and the stack ends are the only heat sink. Real packs also lose heat sideways to cold plates, housings and busbars. This is why the results depend strongly on h. The containment threshold $k_{\mathrm{off}}/L \approx h/3$ should be

read as a statement about the competition between barrier leakage and heat removal, not as a design rule.

- **Lumped kinetics.** One Arrhenius step with a rate ceiling captures onset and total heat release, but not the sequence of real decomposition reactions (SEI breakdown, anode–electrolyte, cathode decomposition, electrolyte combustion, internal short) [1, 10].
- **Missing transport.** Venting, ejecta, gas flow and flames, which often carry much of the energy in real failures [2], are not modelled, and neither are radiation across gaps or contact resistance.
- **Simplified switching.** The idealised switch is instantaneous and reversible. The real-material model adds a response time and one-way switching, but not hysteresis, degradation over cycles or latent heat of the switch itself.
- **Incomplete material data.** Several heat capacities are assumed, the gap pad and PCM are extrapolated far beyond their rated temperatures, the aerogel and ceramic fibre data are for industrial blankets rather than battery sheets, and the switch is described by two published points. Decomposition, melting leakage and combustion of organic barriers are not modelled.
- **Illustrative cell parameters.** The cell and kinetic values are not fitted to data, so temperatures and times are qualitative. The material ranking is more robust than the absolute thicknesses, which scale with the end cooling h.

Natural next steps are to measure $k(T)$ up to runaway temperatures for the leading candidates, especially the thermal switch, including hysteresis and cycling; to fit multi-step kinetics to accelerating-rate calorimetry and calibrate $k_{\max}$ against measured runaway duration; to extend the model to 2D or 3D with lateral cooling, which would reduce the normal-use penalty of insulators; to compare against an independent commercial finite-element model; and to quantify how parameter uncertainty moves the containment boundary.

7 Conclusions

A verified 1D finite-volume model reproduces the barrier conflict between normal-use cooling and abuse containment, and shows that an idealised switchable barrier resolves it. Containment is controlled by the hot-state barrier conductance relative to the heat removal available to the runaway cell. The switch temperature has a broad working window between the normal operating temperature and the onset of self-heating. On the numerical side, mesh-convergence testing exposed a modelling error that a single run would hide: onset-fitted Arrhenius kinetics extrapolated to burning temperatures make propagation times mesh-dependent, and a physically motivated rate ceiling is needed for a converged answer.

Applied to six real barrier materials, the model shows that the conflict is real for static materials. Insulators that contain runaway either raise normal-use temperatures by 28–41 K (aerogel, ceramic fibre) or need a 6 mm layer (mica), and a conductive pad or phase-change composite does not contain runaway at all. A published graphene/microsphere thermal switch is the best candidate: it contains runaway at about 3.5 mm with a normal-use temperature within 1 K of an ideal conductor, a result that holds across all sensitivity cases. Its standing depends on measured high-temperature properties that are not yet available.

Code and data availability. The model, tests and scripts that regenerate every figure and number in this report are available at `https://github.com/adityameenak/switchable-barrier-runaway` under the MIT license. The solver is written in Python using NumPy [28] and SciPy [29].

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References

[1] Xuning Feng, Minggao Ouyang, Xiang Liu, Languang Lu, Yong Xia, and Xiangming He. Thermal runaway mechanism of lithium ion battery for electric vehicles: A review. *Energy Storage Materials*, 10:246–267, 2018. doi: 10.1016/j.ensm.2017.05.013.

[2] Qingsong Wang, Ping Ping, Xuejuan Zhao, Guanquan Chu, Jinhua Sun, and Chunhua Chen. Thermal runaway caused fire and explosion of lithium ion battery. *Journal of Power Sources*, 208:210–224, 2012. doi: 10.1016/j.jpowsour.2012.02.038.

[3] Todd M. Bandhauer, Srinivas Garimella, and Thomas F. Fuller. A Critical Review of Thermal Issues in Lithium-Ion Batteries. *Journal of The Electrochemical Society*, 158(3):R1, 2011. doi: 10.1149/1.3515880.

[4] Joshua Lamb, Christopher J. Orendorff, Leigh Anna M. Steele, and Scott W. Spangler. Failure propagation in multi-cell lithium ion batteries. *Journal of Power Sources*, 283:517–523, 2015. doi: 10.1016/j.jpowsour.2014.10.081.

[5] Carlos F. Lopez, Judith A. Jeevarajan, and Partha P. Mukherjee. Experimental Analysis of Thermal Runaway and Propagation in Lithium-Ion Battery Modules. *Journal of The Electrochemical Society*, 162(9):A1905–A1915, 2015. doi: 10.1149/2.0921509jes.

[6] Jing Wang, Xuning Feng, Yongzheng Yu, Hai Huang, Mengting Zheng, Yunkai Xu, Junxiu Wu, Ying Yang, and Jun Lu. Rapid temperature-responsive thermal regulator for safety management of battery modules. *Nature Energy*, 9:939–946, 2024. doi: 10.1038/s41560-024-01535-5.

[7] Peihang Li, Mingjian Xu, Xiaojie Guo, Pengjun Zheng, Weiping Qu, Chengshuai Xu, Da Liu, WenHui Gan, and Deqiu Zou. Thermal switches for lithium-ion battery thermal management: Principle, performance and application. *Energy Storage Materials*, 81:104481, 2025. doi: 10.1016/j.ensm.2025.104481.

[8] T. D. Hatchard, D. D. MacNeil, A. Basu, and J. R. Dahn. Thermal Model of Cylindrical and Prismatic Lithium-Ion Cells. *Journal of The Electrochemical Society*, 148(7):A755, 2001. doi: 10.1149/1.1377592.

[9] R. Spotnitz and J. Franklin. Abuse behavior of high-power, lithium-ion cells. *Journal of Power Sources*, 113(1):81–100, 2003. doi: 10.1016/s0378-7753(02)00488-3.

[10] Gi-Heon Kim, Ahmad Pesaran, and Robert Spotnitz. A three-dimensional thermal abuse model for lithium-ion cells. *Journal of Power Sources*, 170(2):476–489, 2007. doi: 10.1016/j.jpowsour .2007.04.018.

[11] Carlos F. Lopez, Judith A. Jeevarajan, and Partha P. Mukherjee. Characterization of Lithium-Ion Battery Thermal Abuse Behavior Using Experimental and Computational Analysis. *Journal of The Electrochemical Society*, 162(10):A2163–A2173, 2015. doi: 10.1149/2.0751510jes.

[12] Xuning Feng, Xiangming He, Minggao Ouyang, Languang Lu, Peng Wu, Christian Kulp, and Stefan Prasser. Thermal runaway propagation model for designing a safer battery pack with 25 Ah $LiNi_xCo_yMn_zO_2$ large format lithium ion battery. *Applied Energy*, 154:74–91, 2015. doi: 10.1016/j.apenergy.2015.04.118.

[13] Xuning Feng, Languang Lu, Minggao Ouyang, Jiangqiu Li, and Xiangming He. A 3D thermal runaway propagation model for a large format lithium ion battery module. *Energy*, 115:194–208, 2016. doi: 10.1016/j.energy.2016.08.094.

[14] Zhiguo Tang, Anqi Song, Shoucheng Wang, Jianping Cheng, and Changfa Tao. Numerical Analysis of Heat Transfer Mechanism of Thermal Runaway Propagation for Cylindrical Lithium-ion Cells in Battery Module. *Energies*, 13(4):1010, 2020. doi: 10.3390/en13041010.

[15] Stephen Wilke, Ben Schweitzer, Siddique Khateeb, and Said Al-Hallaj. Preventing thermal runaway propagation in lithium ion battery packs using a phase change composite material: An experimental study. *Journal of Power Sources*, 340:51–59, 2017. doi: 10.1016/j.jpowsour.2016.11.018.

[16] Shaw Kang Wong, Kuijie Li, Xinyu Rui, Liyun Fan, Minggao Ouyang, and Xuning Feng. Mitigating thermal runaway propagation in high specific energy lithium-ion battery modules through nanofiber aerogel composite material. *Energy*, 307:132353, 2024. doi: 10.1016/j.energy.2024.132353.

[17] Fabian Menz, Bruno Bausch, Joaquín Klee Barillas, Olaf Böse, Michael A. Danzer, and Markus Hölzle. Preventing thermal runaway propagation in lithium-ion batteries: Model-based optimization of interstitial heat-absorbing thermal barriers. *Journal of Power Sources*, 584:233578, 2023. doi: 10.1016/j.jpowsour.2023.233578.

[18] Yang Li, Yong-Yan Xie, Yu-Jie Zhang, et al. Recent Advances in Polymer-Based Thermal Barrier Materials for Mitigating Thermal Runaway Propagation in Lithium-Ion Batteries. *Polymers*, 18(7):801, 2026. doi: 10.3390/polym18070801.

[19] Geoff Wehmeyer, Tomohide Yabuki, Christian Monachon, Junqiao Wu, and Chris Dames. Thermal diodes, regulators, and switches: Physical mechanisms and potential applications. *Applied Physics Reviews*, 4(4):041304, 2017. doi: 10.1063/1.5001072.

[20] Aspen Aerogels. Pyrogel XTE product data sheet, Rev 4.1. https://www.aerogel.com/wp-content/uploads/2021/06/Pyrogel_XTE_Datasheet_English_v4.1.pdf, 2022. Accessed November 2026.

[21] The Gund Company. Mica P – flexible silicone bonded mica paper laminate, technical data sheet. https://thegundcompany.com/wp-content/uploads/2025/04/Mica-P-Flexible-Silicone-Bonded-Mica-Paper-Laminate-by-The-Gund-Co-TDS.pdf, 2026. Accessed November 2026.

[22] Morgan Advanced Materials. Refractory ceramic fiber blanket, datasheet 5-14-205. https://www.pacorinc.com/pdf/refractory_ceramic_fiber_blanket-datasheet.pdf, 2016. Accessed November 2026.

[23] Henkel. BERGQUIST GAP PAD TGP 3000 product information. https://www.henkel-adhesives.com/ca/en/product/thermal-gap-pad-materials/bergquist_gap_padtgp3000.html, 2026. Values from a search summary; page not retrieved directly. Accessed November 2026.

[24] Yilin Zhao, Shuhui Huang, Zhaoguo Jin, Zhongnan Xie, Hong Guo, and Haofeng Xie. Thermally conductive shape-stabilized phase change materials enabled by paraffin wax and nanoporous structural expanded graphite. *Nanomaterials*, 15(2):110, 2025. doi: 10.3390/nano15020110.

[25] Suhas V. Patankar. *Numerical Heat Transfer and Fluid Flow*. Hemisphere Publishing, Washington, DC, 1980.

[26] Gilbert Strang. On the construction and comparison of difference schemes. *SIAM Journal on Numerical Analysis*, 5(3):506–517, 1968. doi: 10.1137/0705041.

[27] Ernst Hairer and Gerhard Wanner. *Solving Ordinary Differential Equations II*. Springer Series in Computational Mathematics. Springer Berlin Heidelberg, 2 edition, 1996. doi: 10.1007/978-3-642-05221-7.

[28] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.

[29] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2.